

The Least Residue-Choice Incompatibility Layer for Finite Survivor Sets: A Complete Classification

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Abstract

For a finite residue profile with one forbidden class modulo each integer $2, \dots, n$, a survivor set may pass every pointwise test for extension to level $n + 1$ while the required residue choices are mutually incompatible. We determine the least level at which this occurs. It is $n = 24$, and the complete exception family is

$$\{\{2\} \cup T : T \subseteq \{8, 14, 20\}\}.$$

The result is certified by exact exhaustive generation through level 24, a separate exact covering-CSP computation, and an implementation-independent set-based replay. All eight exceptions have the same transparent obstruction: the omitted points 4 and 6 force incompatible values of the residue chosen modulo 3. This is a finite classification only; the motivating asymptotic question concerning the ratio of extendible to all survivor sets remains unresolved.

1 Finite survivor sets and the local test

For $n \geq 1$, put $X_n = \{2, 3, \dots, n + 1\}$ and

$$R_n = \prod_{k=2}^n \mathbb{Z}/k\mathbb{Z}.$$

The empty product R_1 contains one profile. For $r = (r_2, \dots, r_n) \in R_n$, define

$$S_n(r) = \{m \in X_n : [m]_k \neq r_k \text{ for every } 2 \leq k < m\}, \quad \mathcal{F}_n = \{S_n(r) : r \in R_n\}.$$

Write $N(n) = |\mathcal{F}_n|$. A set $A \in \mathcal{F}_n$ is *extendible* if some profile realizing A also satisfies $r_k \neq [n + 2]_k$ for every $2 \leq k \leq n$; denote the extendible subfamily by $\mathcal{E}_n$ and its size by $E(n)$. Equivalently,

$$A \in \mathcal{E}_n \iff A \cup \{n + 2\} \in \mathcal{F}_{n+1}.$$

These objects, their exact covering characterization, and a dynamic recurrence for $\mathcal{F}_n$ were introduced by Raso and Venturi [1]. We recall the required definitions so that the finite classification below is self-contained. For $A \subseteq X_n$ and $2 \leq k \leq n$, let

$$\Omega_k^+(A) = \mathbb{Z}/k\mathbb{Z} \setminus (\{[x]_k : x \in A, x > k\} \cup \{[n + 2]_k\}).$$

For $m \in X_n \setminus A$, set

$$W_A(m) = \{k : 2 \leq k < m, [m]_k \in \Omega_k^+(A)\}.$$

Let $L_n(A)$ mean that every $\Omega_k^+(A)$ is nonempty and every omitted point m has $W_A(m) \neq \emptyset$. Finally, define

$$\mathcal{G}_n = \{A \in \mathcal{F}_n \setminus \mathcal{E}_n : L_n(A)\}.$$

Thus $\mathcal{G}_n$ consists exactly of survivor sets that pass both kinds of local nonemptiness test but fail global compatibility.

For $a \in \Omega_k^+(A)$, let

$$C_{k,a}(A) = \{m \in X_n \setminus A : m > k, [m]_k = a\}.$$

The following elementary formulation is the finite CSP used below.

Lemma 1 (Exact covering criterion). *If $A \in \mathcal{F}_n$, then $A \in \mathcal{E}_n$ if and only if one can choose $a_k \in \Omega_k^+(A)$ for $2 \leq k \leq n$ such that*

$$X_n \setminus A \subseteq \bigcup_{k=2}^n C_{k,a_k}(A).$$

Equivalently, every omitted m contributes the clause

$$\bigvee_{k \in W_A(m)} (a_k = [m]_k).$$

Proof. If a profile extends A , its coordinates lie in the displayed domains: they eliminate neither a survivor nor the prospective point $n+2$. Because the profile realizes A , every omitted point is in one of its selected blocks. Conversely, domain membership protects $A \cup \{n+2\}$, while the covering condition eliminates every point outside A . The selected residues therefore form an extending profile. $\square$

The nearest source result records that the pointwise witness condition need not be sufficient because several points can require incompatible choices at one modulus [1, Remark 34]. The finite delta established here is the least layer, its complete literal exception family, and explicit conflict cores. Since that source is a recent first version and no exhaustive priority search is asserted here, this comparison is deliberately limited to the cited paper and no broader priority claim is made.

For a bibliography-designated methodological comparison only, Zeng's work [2] is titled *A Strict Certified Search Extension for Erdős Problem 647*. We cite it only as an adjacent example of a certification-oriented search extension attached to a numbered problem. This title-level comparison uses the user-designated bibliography record; that work is not a source for the survivor-set definitions or theorem, and no result from it is used in the proof below.

2 Result and exact certificate

Theorem 2. *The least integer n for which $\mathcal{G}_n$ is nonempty is*

$$n_* = 24.$$

At that level the complete exception family is

$$\mathcal{G}_{24} = \{\{2\} \cup T : T \subseteq \{8, 14, 20\}\}.$$

In particular, it consists of exactly eight sets.

We first describe the exhaustive part of the proof. For fixed n , modulus k , and $a \in \mathbb{Z}/k\mathbb{Z}$, define the elimination block

$$B_{k,a}^{(n)} = \{m \in X_n : k < m, [m]_k = a\}.$$

Starting from $\mathcal{U}_{1,n} = \{\emptyset\}$, set

$$\mathcal{U}_{j,n} = \{U \cup B_{j,a}^{(n)} : U \in \mathcal{U}_{j-1,n}, a \in \mathbb{Z}/j\mathbb{Z}\} \quad (2 \leq j \leq n), \quad (1)$$

with exact set deduplication at every stage.

Lemma 3 (Completeness of the generator). *For every $n \geq 1$,*

$$\mathcal{F}_n = \{X_n \setminus U : U \in \mathcal{U}_{n,n}\}.$$

Proof. After coordinates $2, \dots, j$ have been chosen, the eliminated set is exactly $B_{2,r_2}^{(n)} \cup \dots \cup B_{j,r_j}^{(n)}$. Induction on j shows that (1) contains all and only these partial unions. Taking complements after the last coordinate gives precisely all survivor sets. $\square$

The implementation stores subsets as integer bit masks, so (1) uses bitwise union and exact integer equality only. For every generated A , the CSP was constructed directly from $\Omega_k^+(A)$ and $C_{k,a}(A)$, rather than being inferred from recurrence (1). Its exact dynamic program begins with the empty covered set and, successively for $k = 2, \dots, n$, adjoins each block $C_{k,a}(A)$ with $a \in \Omega_k^+(A)$, again deduplicating exact coverage sets. By induction on k , its states are precisely the coverages attainable by one allowed value for every processed modulus. Lemma 1 then decides extendibility exactly.

The complete output is shown in Table 1. Here the third column is the number accepted by the covering CSP, the fourth is the number satisfying L_n , and the final column is $|\mathcal{G}_n|$. Thus this is a full-family calculation, not a sampling or floating-point test.

As a separate replay, the survivor families were regenerated with immutable sets rather than bit masks. Extendibility was then decided by attainable coverage-union states, and the complete survivor, extendible, and locally feasible subfamilies were compared set-for-set with the first computation. The replay also generated $|\mathcal{F}_{25}| = 666$ and checked, for each of the 553 sets $A \in \mathcal{F}_{24}$, that the CSP decision agrees with $A \cup \{26\} \in \mathcal{F}_{25}$. Both sides contain 113 sets. These independent representations reproduce every row of Table 1 and the same eight exceptions.

It remains to expose the membership, local feasibility, and incompatibility of those eight sets without treating the enumeration as a black box.

Proposition 4 (Explicit realization). *Every $A = \{2\} \cup T$ with $T \subseteq \{8, 14, 20\}$ belongs to $\mathcal{F}_{24}$.*

Proof. In every row of Table 2, take

$$r_2 = r_3 = r_5 = 1, \quad r_6 = r_8 = r_9 = r_{11} = r_{12} = 0, \quad r_{14} = \dots = r_{24} = 0,$$

and use the four displayed coordinates. Direct substitution in the definition of S_{24} gives $S_{24}(r) = \{2\} \cup T$. $\square$

Proposition 5 (Uniform local certificate). *Every set in Proposition 4 satisfies L_{24} .*

Table 1: Exact exhaustive enumeration through the least incompatibility layer.

n	$N(n)$	$ \mathcal{E}_n $	$ \{A \in \mathcal{F}_n : L_n(A)\} $	$ \mathcal{G}_n $
1	1	1	1	0
2	2	1	1	0
3	3	1	1	0
4	4	2	2	0
5	6	2	2	0
6	8	3	3	0
7	11	3	3	0
8	14	3	3	0
9	17	5	5	0
10	22	8	8	0
11	30	8	8	0
12	38	12	12	0
13	50	13	13	0
14	63	13	13	0
15	76	13	13	0
16	89	28	28	0
17	117	30	30	0
18	147	45	45	0
19	192	45	45	0
20	237	51	51	0
21	288	61	61	0
22	349	102	102	0
23	451	102	102	0
24	553	113	121	8

Proof. For every one of the eight sets, the following choices exhibit a member of each residue domain:

k	2	3	4	5	6	7	8	9	10	11	12	13	14-24
d_k	1	0	1	2	0	2	0	0	1	0	0	2	0.

Thus all domains are nonempty. Every omitted odd point is witnessed by $(k, [m]_k) = (2, 1)$. Every omitted even point congruent to 0 or 1 modulo 3 is witnessed respectively by $(3, 0)$ or $(3, 1)$. The only remaining possible omitted points are 8, 14, 20; they are witnessed respectively by $(5, 3)$, $(5, 4)$, $(5, 0)$. These three residues are distinct and none is $[26]_5 = 1$, so whenever one of those points is omitted its indicated residue is allowed. Every omitted point therefore has a witness. $\square$

Proposition 6 (A common two-clause core). *No set in Proposition 4 is extendible.*

Proof. For every such set A , all its elements larger than 2 are even, and the prospective point is 26. Hence $\Omega_2^+(A) = \{1\}$. Moreover, each of 8, 14, 20, 26 is congruent to 2 modulo 3, so $\Omega_3^+(A) = \{0, 1\}$.

The point 4 is omitted. Modulus 2 offers residue 0, which is forbidden, while modulus 3 offers residue 1. Therefore $W_A(4) = \{3\}$, and the clause for 4 forces $a_3 = 1$. The point 6 is also omitted. Its residues at moduli 2, 4, 5 are forbidden: they are respectively 0, $2 = [26]_4$, and $1 = [26]_5$. Its only allowed witness is therefore modulus 3, so $W_A(6) = \{3\}$ and the clause for 6 forces $a_3 = 0$. These requirements are incompatible. Lemma 1 proves $A \notin \mathcal{E}_{24}$. $\square$

Proof of Theorem 2. Lemmas 3 and 1, together with the exhaustive exact calculation in Table 1, show that $\mathcal{G}_n$ is empty for $1 \leq n < 24$ and has exactly eight members at $n = 24$. The set-for-set

Table 2: The varying coordinates of profiles realizing all eight sets.

T	r_4	r_7	r_{10}	r_{13}
$\emptyset$	0	0	0	0
$\{8\}$	1	0	0	0
$\{14\}$	0	1	0	0
$\{20\}$	1	1	1	1
$\{8, 14\}$	1	2	0	0
$\{8, 20\}$	1	0	1	0
$\{14, 20\}$	1	1	1	0
$\{8, 14, 20\}$	1	2	1	0

output at level 24 is exactly the family displayed in the theorem. Propositions 4–6 give explicit human-readable certificates that every displayed set is a survivor, passes all local tests, and fails the global CSP. This proves both minimality and the complete literal classification. $\square$

3 Scope and reproducibility

The theorem identifies a finite boundary between pointwise witnesses and a globally consistent residue profile. It neither estimates nor resolves the asymptotic question $E(n)/N(n) \rightarrow 0$. It also makes no claim about levels above 24.

The supplementary certificate consists of two exact generators, a direct CSP solver, serialized full-family data through $n = 24$, and an independent replay. The principal verification commands are

```
python evidence/find_first_incompatibility.py --max-n 80 \
    --output evidence/first_incompatibility.json
python evidence/independent_audit.py
```

where the first program stops when it proves the least nonempty layer, so its actual coverage is exactly $1 \leq n \leq 24$. The archived execution uses a fixed project interpreter; the displayed commands emphasize the script-level reproducibility rather than a machine-specific path. All mathematical decisions use finite integer, residue, bit-set, or immutable-set operations; no floating-point tolerance, heuristic solver, or probabilistic completeness claim is involved.

An OpenAI Codex language model assisted with implementation, independent replay design, proof auditing, and manuscript drafting. That assistance is not used as mathematical evidence: the stated result rests on the explicit proofs, exhaustive certificates, and separately implemented exact replays described above.

References

- [1] Mario Raso and Daniele Venturi. Counting survivor sets: Exponential equivalence with prime-admissible sets, 2026. arXiv:2609.08528v1, 8 September 2026.
- [2] Zijian Zeng. A Strict Certified Search Extension for Erdős Problem 647. 2026.